CHAPTER1-DAA LAB EXPERIMENTS

1. First Palindromic String

```python
def firstPalindrome(words):
    for word in words:
        if word == word[::-1]:
            return word
    return ""
words = ["abc", "car", "ada", "racecar", "cool"]
print(firstPalindrome(words))
```

```
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 1"
ada
```

2. Count Common Elements

```python
def findAnswer(nums1, nums2):
    set1 = set(nums1)
    set2 = set(nums2)
    answer1 = sum(1 for x in nums1 if x in set2)
    answer2 = sum(1 for x in nums2 if x in set1)
    return [answer1, answer2]
nums1 = [2, 3, 2]
nums2 = [1, 2]
print(findAnswer(nums1, nums2))
```

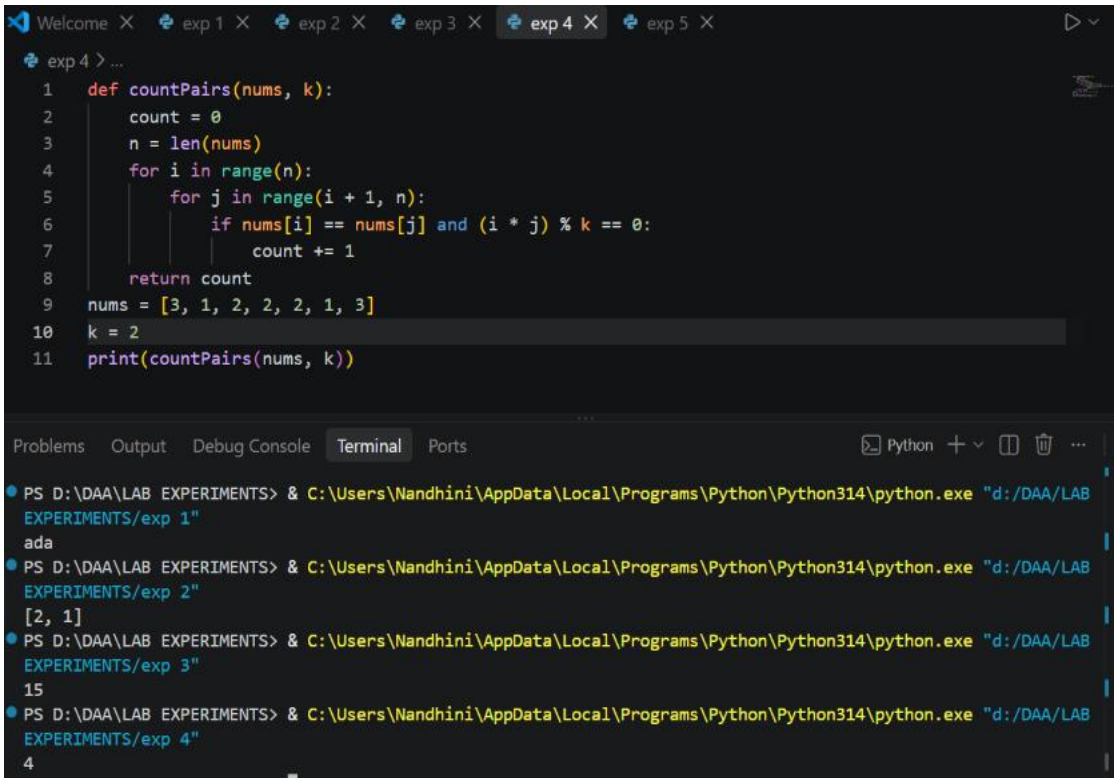
```
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 1"
ada
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 2"
[2, 1]
PS D:\DAA\LAB EXPERIMENTS>
```

3. Sum of Squares of Distinct Counts

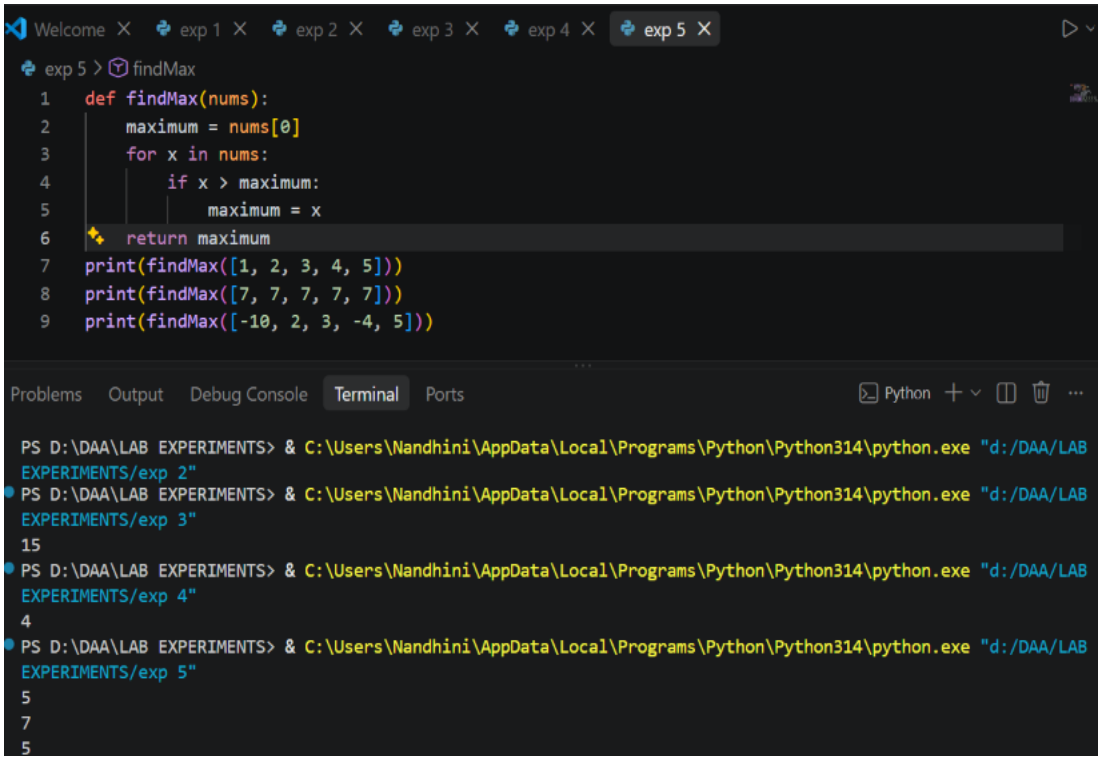
```python
def sumCounts(nums):
    total = 0
    n = len(nums)
    for i in range(n):
        distinct = set()
        for j in range(i, n):
            distinct.add(nums[j])
            count = len(distinct)
            total += count * count
    return total
nums = [1, 2, 1]
print(sumCounts(nums))
```

```
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 1"
ada
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 2"
[2, 1]
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 3"
15
PS D:\DAA\LAB EXPERIMENTS>
```

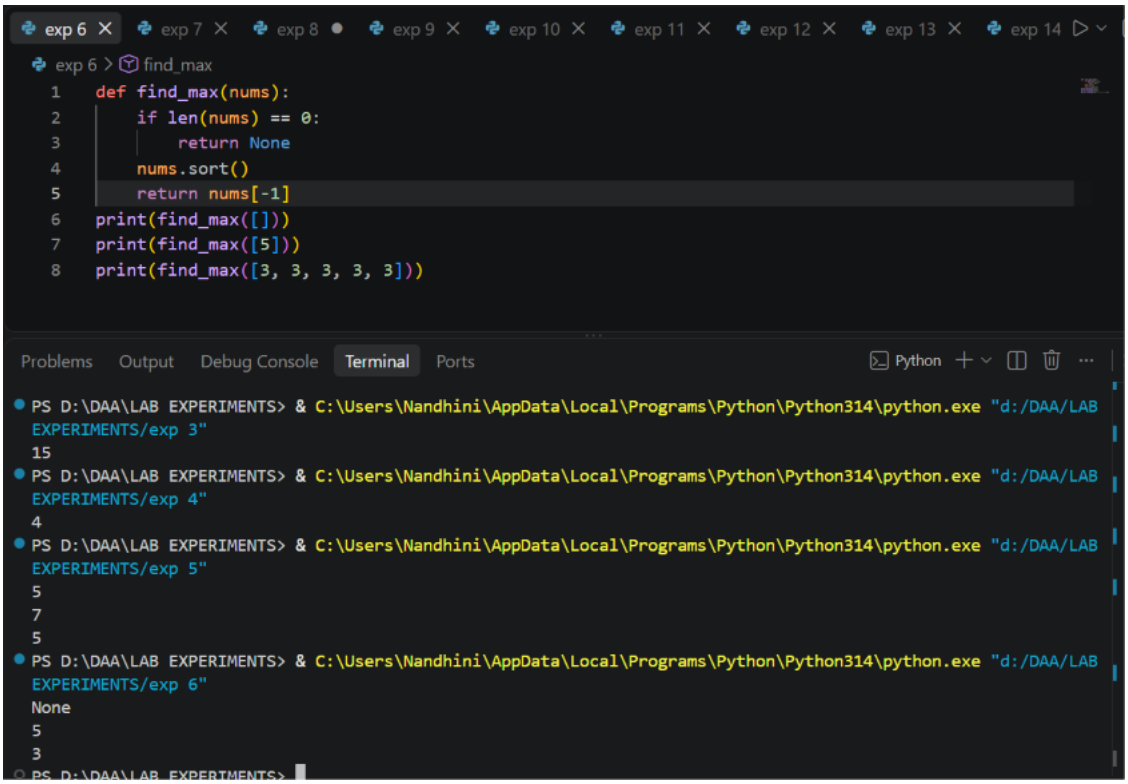
4. Count Valid Pairs



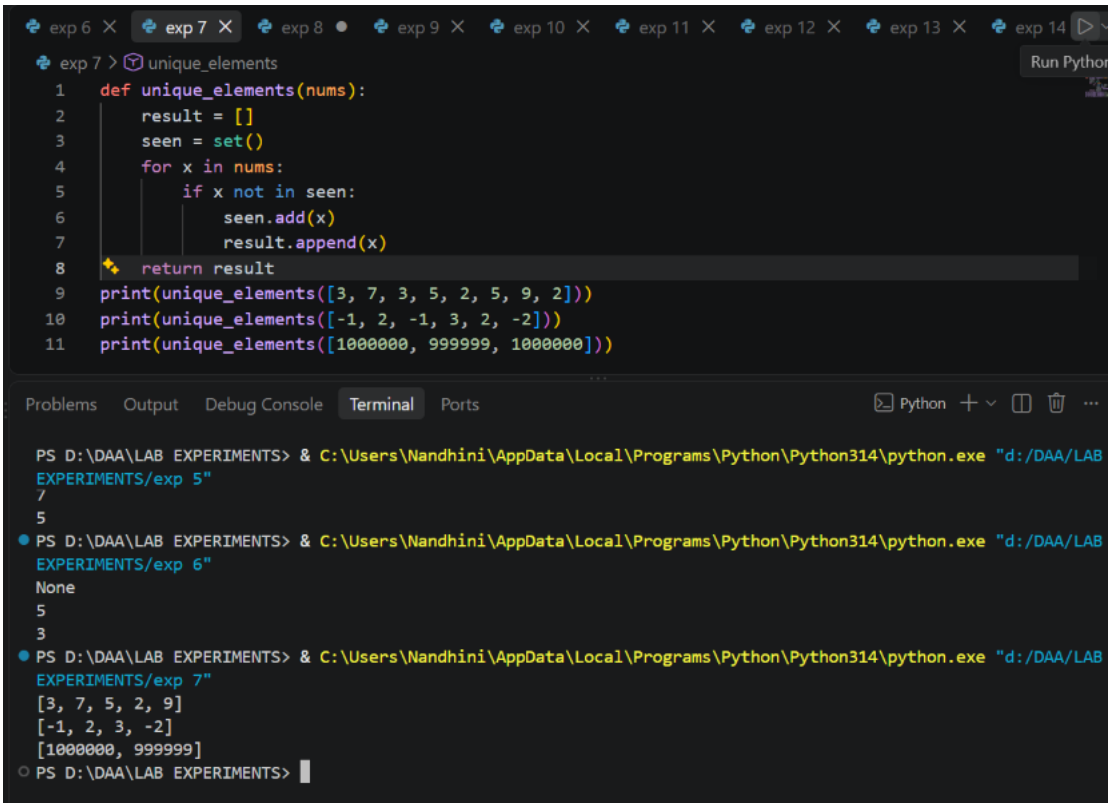
5. Least Time Complexity



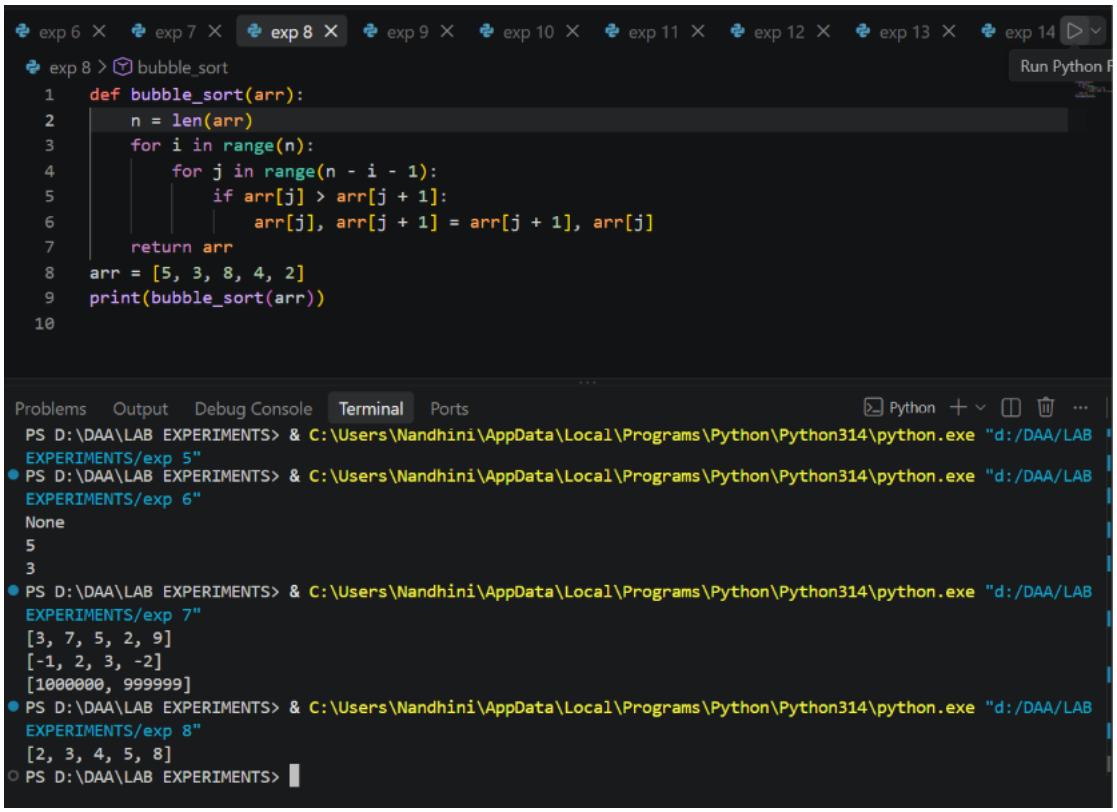
6. Sort and Find Maximum



7. Unique Elements



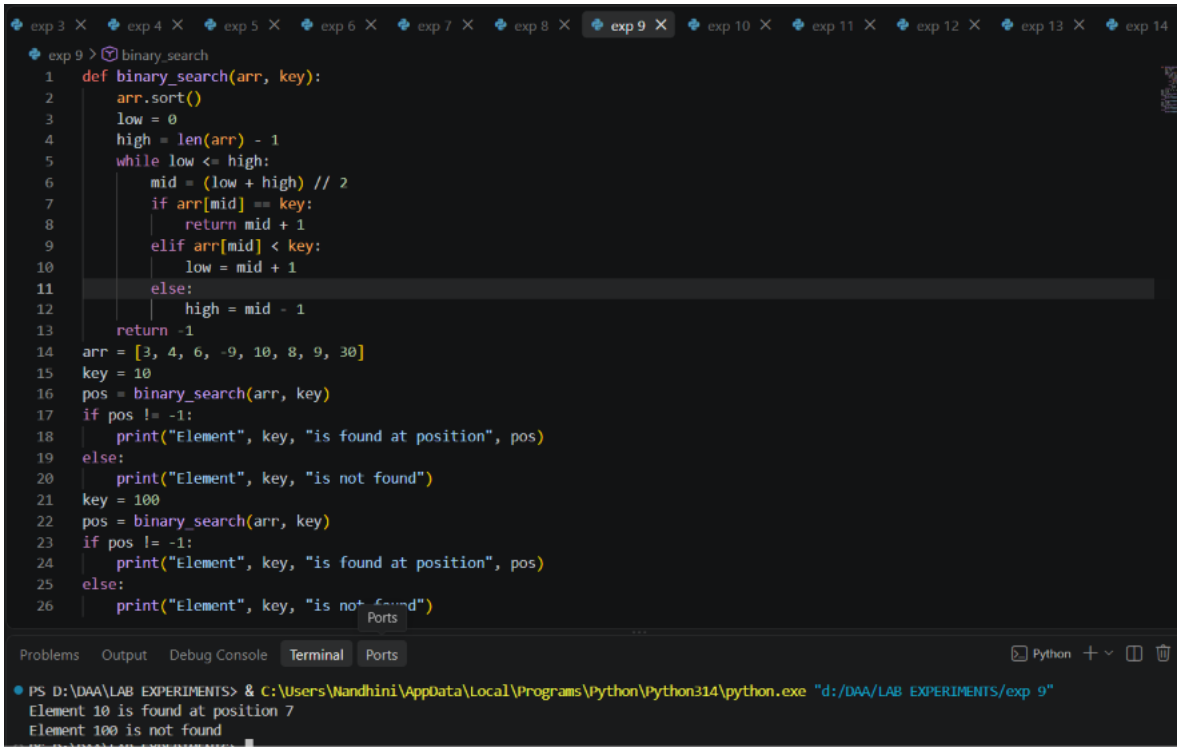
8. Bubble Sort



```
exp 6 | exp 7 | exp 8 | exp 9 | exp 10 | exp 11 | exp 12 | exp 13 | exp 14
exp 8 > bubble_sort
1  def bubble_sort(arr):
2      n = len(arr)
3      for i in range(n):
4          for j in range(n - i - 1):
5              if arr[j] > arr[j + 1]:
6                  arr[j], arr[j + 1] = arr[j + 1], arr[j]
7      return arr
8  arr = [5, 3, 8, 4, 2]
9  print(bubble_sort(arr))
10

Problems  Output  Debug Console  Terminal  Ports
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 5"
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 6"
None
5
3
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 7"
[3, 7, 5, 2, 9]
[-1, 2, 3, -2]
[1000000, 999999]
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 8"
[2, 3, 4, 5, 8]
PS D:\DAA\LAB EXPERIMENTS>
```

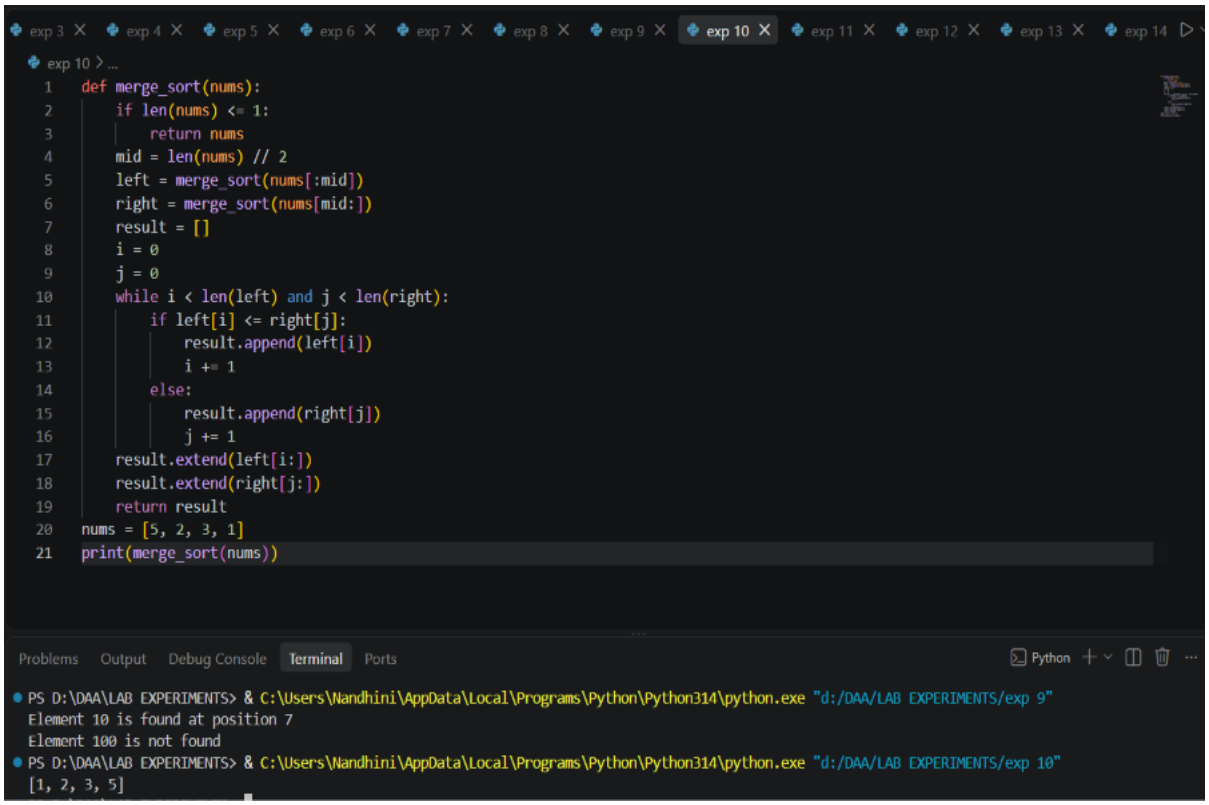
9. Binary Search



```
exp 3 | exp 4 | exp 5 | exp 6 | exp 7 | exp 8 | exp 9 | exp 10 | exp 11 | exp 12 | exp 13 | exp 14
exp 9 > binary_search
1  def binary_search(arr, key):
2      arr.sort()
3      low = 0
4      high = len(arr) - 1
5      while low <= high:
6          mid = (low + high) // 2
7          if arr[mid] == key:
8              return mid + 1
9          elif arr[mid] < key:
10             low = mid + 1
11         else:
12             high = mid - 1
13     return -1
14 arr = [3, 4, 6, -9, 10, 8, 9, 30]
15 key = 10
16 pos = binary_search(arr, key)
17 if pos != -1:
18     print("Element", key, "is found at position", pos)
19 else:
20     print("Element", key, "is not found")
21 key = 100
22 pos = binary_search(arr, key)
23 if pos != -1:
24     print("Element", key, "is found at position", pos)
25 else:
26     print("Element", key, "is not found")

Problems  Output  Debug Console  Terminal  Ports
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 9"
Element 10 is found at position 7
Element 100 is not found
```

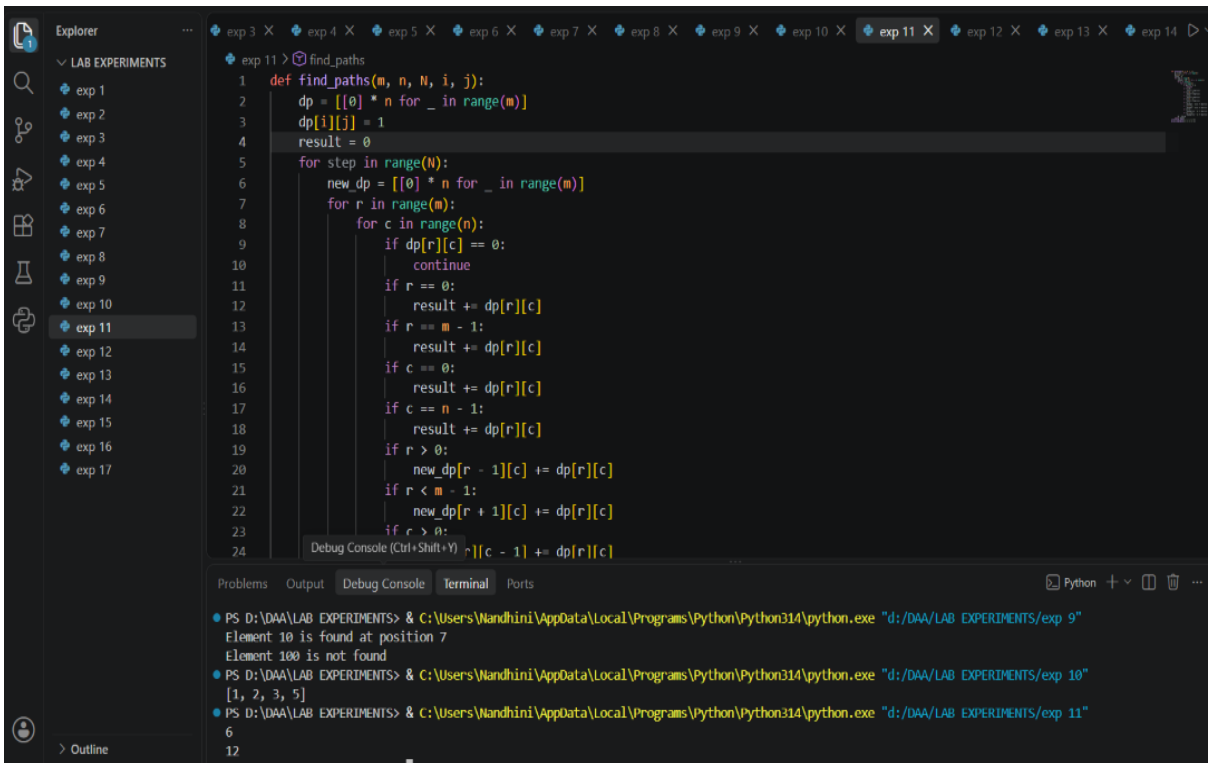
10. Sort Array Without Built-in Functions — Merge Sort



```
def merge_sort(nums):
    if len(nums) <= 1:
        return nums
    mid = len(nums) // 2
    left = merge_sort(nums[:mid])
    right = merge_sort(nums[mid:])
    result = []
    i = 0
    j = 0
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1
    result.extend(left[i:])
    result.extend(right[j:])
    return result
nums = [5, 2, 3, 1]
print(merge_sort(nums))
```

```
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 9"
Element 10 is found at position 7
Element 100 is not found
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 10"
[1, 2, 3, 5]
```

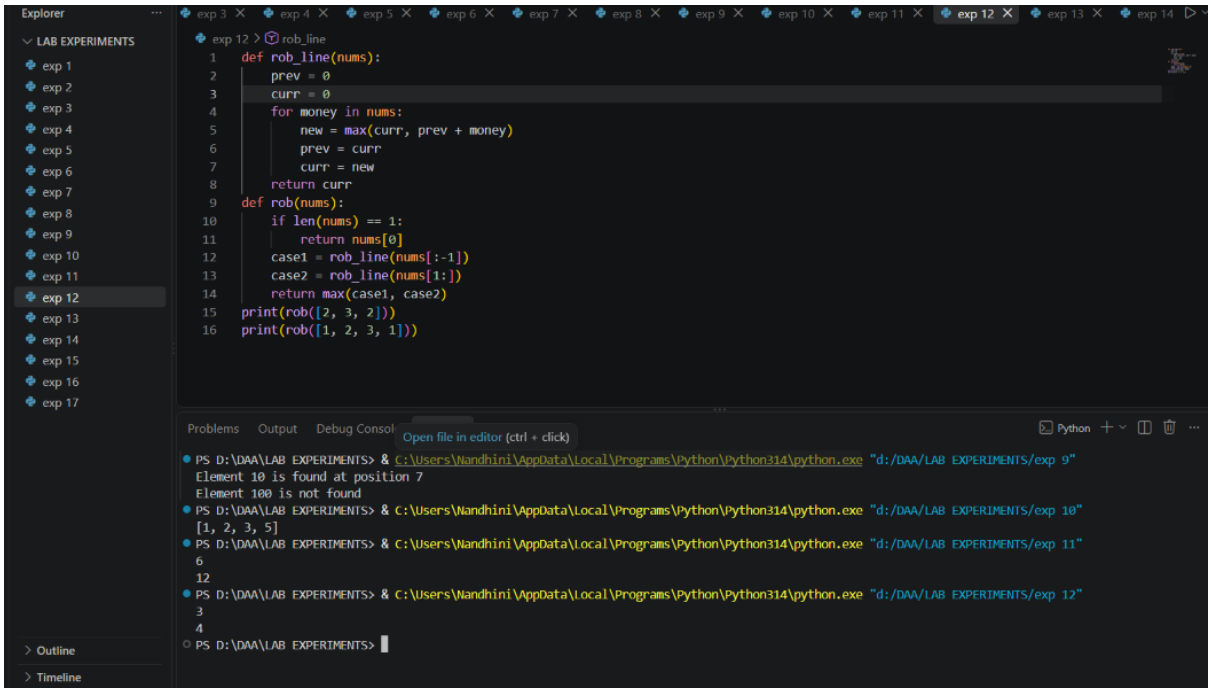
11. Ball Out of Grid



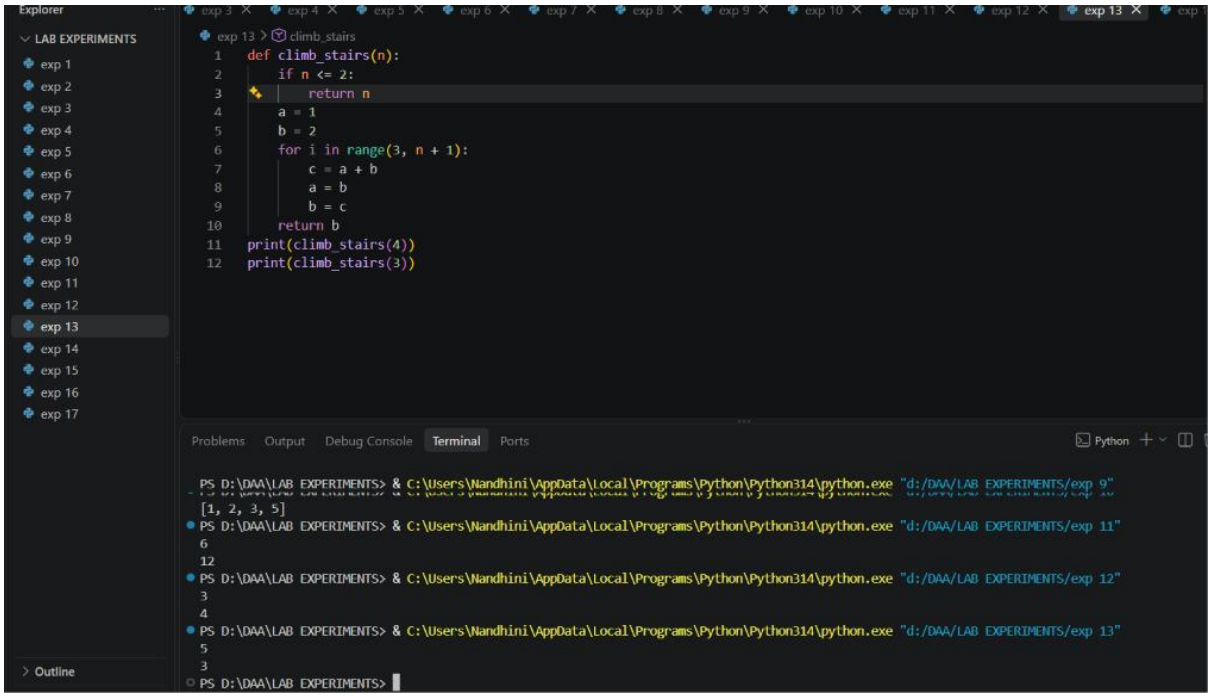
```
def find_paths(m, n, N, i, j):
    dp = [[0] * n for _ in range(m)]
    dp[i][j] = 1
    result = 0
    for step in range(N):
        new_dp = [[0] * n for _ in range(m)]
        for r in range(m):
            for c in range(n):
                if dp[r][c] == 0:
                    continue
                if r == 0:
                    result += dp[r][c]
                if r == m - 1:
                    result += dp[r][c]
                if c == 0:
                    result += dp[r][c]
                if c == n - 1:
                    result += dp[r][c]
                if r > 0:
                    new_dp[r - 1][c] += dp[r][c]
                if r < m - 1:
                    new_dp[r + 1][c] += dp[r][c]
                if c > 0:
```

```
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 9"
Element 10 is found at position 7
Element 100 is not found
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 10"
[1, 2, 3, 5]
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 11"
6
12
```

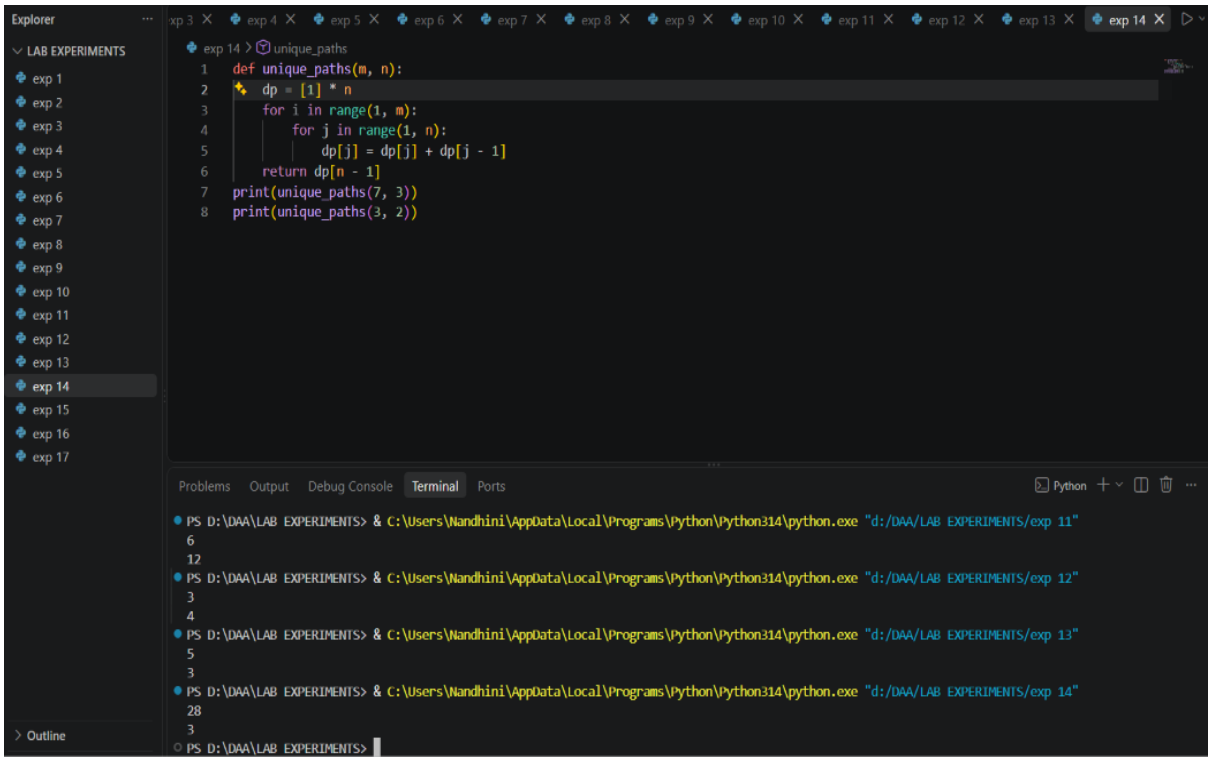
12. House Robber II



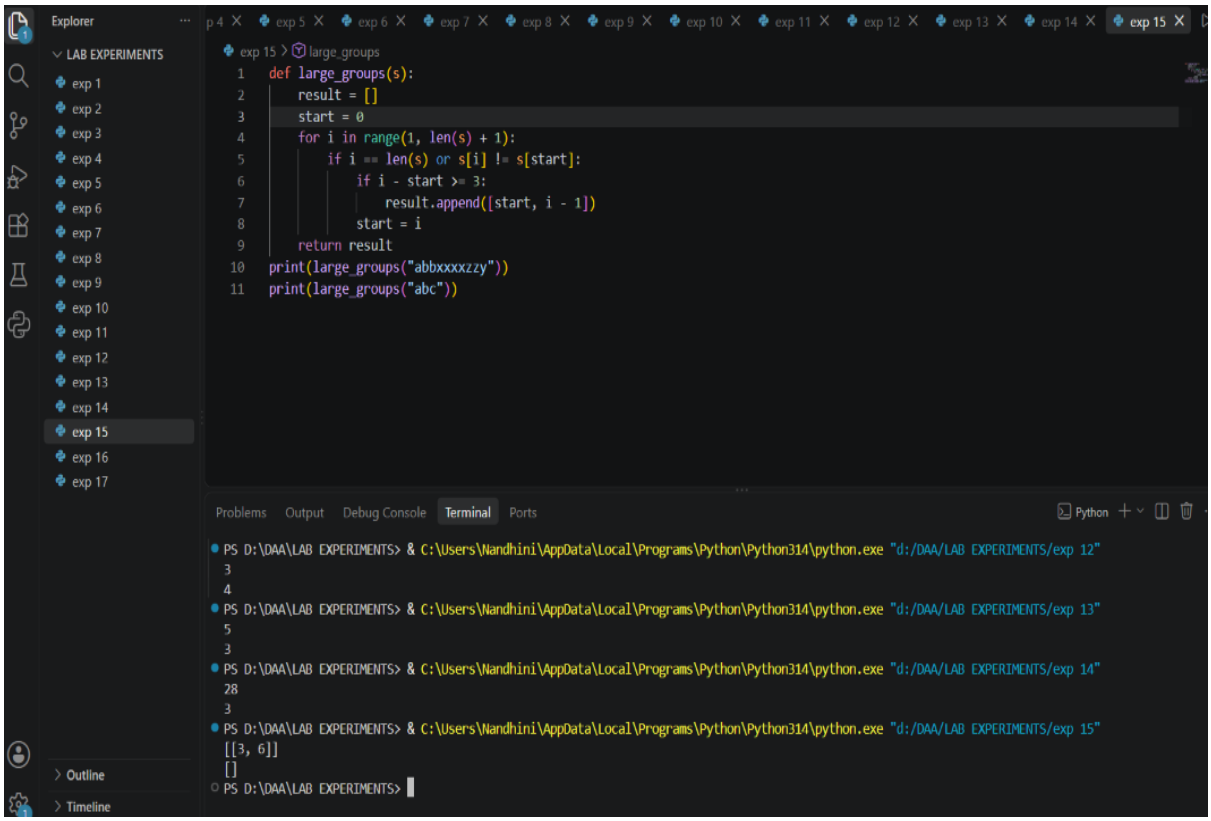
13. Climbing Stairs



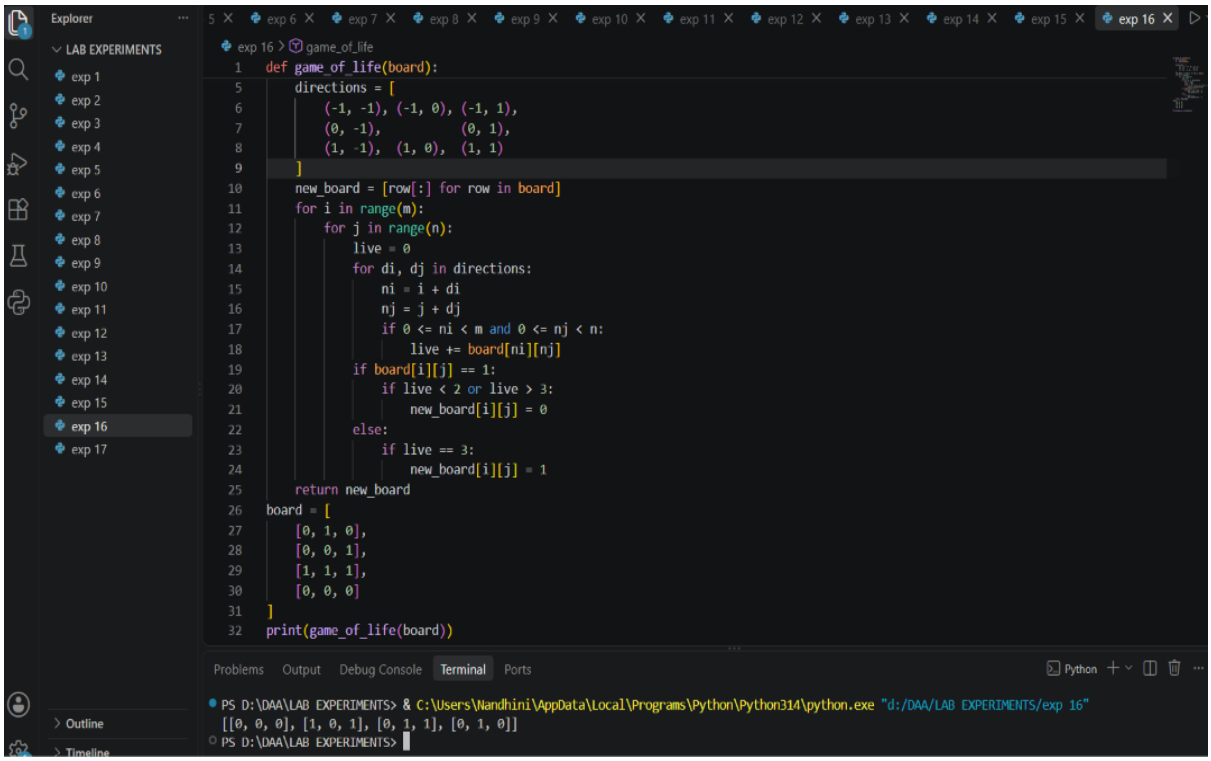
14. Unique Paths



15. Large Groups in a String



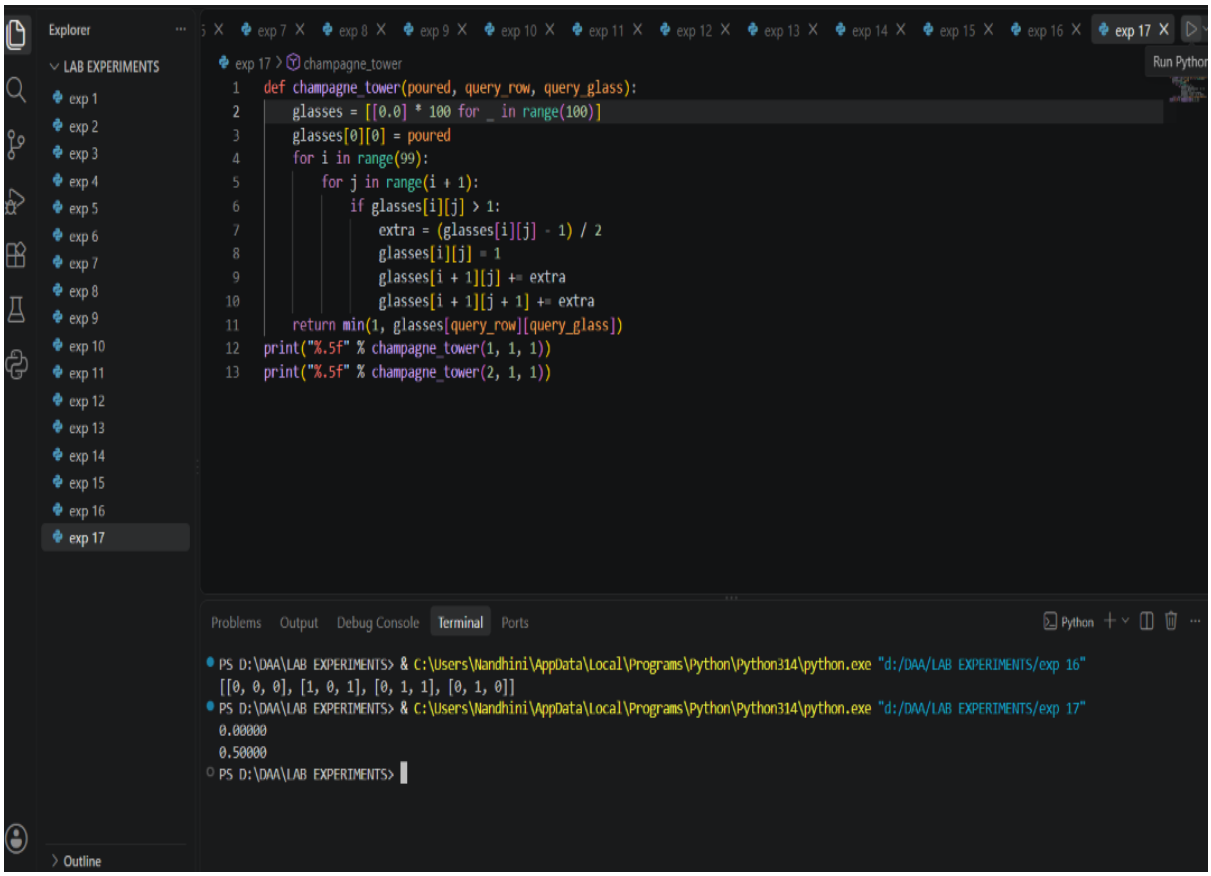
16. Game of Life



```
def game_of_life(board):
    directions = [
        (-1, -1), (-1, 0), (-1, 1),
        (0, -1),           (0, 1),
        (1, -1),  (1, 0),  (1, 1)
    ]
    new_board = [row[:] for row in board]
    for i in range(m):
        for j in range(n):
            live = 0
            for di, dj in directions:
                ni = i + di
                nj = j + dj
                if 0 <= ni < m and 0 <= nj < n:
                    live += board[ni][nj]
            if board[i][j] == 1:
                if live < 2 or live > 3:
                    new_board[i][j] = 0
            else:
                if live == 3:
                    new_board[i][j] = 1
    return new_board
board = [
    [0, 1, 0],
    [0, 0, 1],
    [1, 1, 1],
    [0, 0, 0]
]
print(game_of_life(board))
```

```
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 16"
[[0, 0, 0], [1, 0, 1], [0, 1, 1], [0, 1, 0]]
PS D:\DAA\LAB EXPERIMENTS>
```

17. Champagne Tower



```
def champagne_tower(poured, query_row, query_glass):
    glasses = [[0.0] * 100 for _ in range(100)]
    glasses[0][0] = poured
    for i in range(99):
        for j in range(i + 1):
            if glasses[i][j] > 1:
                extra = (glasses[i][j] - 1) / 2
                glasses[i][j] = 1
                glasses[i + 1][j] += extra
                glasses[i + 1][j + 1] += extra
    return min(1, glasses[query_row][query_glass])
print("%.5f" % champagne_tower(1, 1, 1))
print("%.5f" % champagne_tower(2, 1, 1))
```

```
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 16"
[[0, 0, 0], [1, 0, 1], [0, 1, 1], [0, 1, 0]]
PS D:\DAA\LAB EXPERIMENTS> & C:\Users\Nandhini\AppData\Local\Programs\Python\Python314\python.exe "d:/DAA/LAB EXPERIMENTS/exp 17"
0.00000
0.50000
PS D:\DAA\LAB EXPERIMENTS>
```